LOGISTICS RISK PLATFORM · MCA SEMESTER 3 PROJECT

ChainShield

Transportation System

Real-Time Disruption Prediction & Route Optimisation for Logistics

Group No. 59

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THE PROBLEM

Disruptions are found only after it's too late

1

Reactive monitoring

Managers learn about a delay only after a customer reports a missed delivery.

2

Siloed signals

Weather, congestion, cargo risk, and delay history are never fused into one score.

3

Ad-hoc rerouting

When a disruption is found, the reroute decision relies on guesswork, not data.

OBJECTIVE

Catch disruptions before they cascade

- Continuously score every active shipment's disruption risk
- Auto-flag disruptions the instant a risk threshold is crossed
- Instantly propose ranked, optimised route alternatives
- Notify managers, then field operators, the moment a decision is made
- Give each role — admin, manager, operator — its own focused dashboard

GOAL

Reactive

→

Preemptive

PROPOSED SOLUTION

How ChainShield works

1

MONITOR

Every active shipment is scored on a recurring interval using live weather, congestion, cargo, and history signals.

2

DETECT

Reaching MEDIUM risk auto-creates a disruption and alerts managers live — escalated if it reaches HIGH.

3

REROUTE

3–5 ranked options (hold, hub reroute, mode switch) with time, ₹ cost and residual-risk trade-offs.

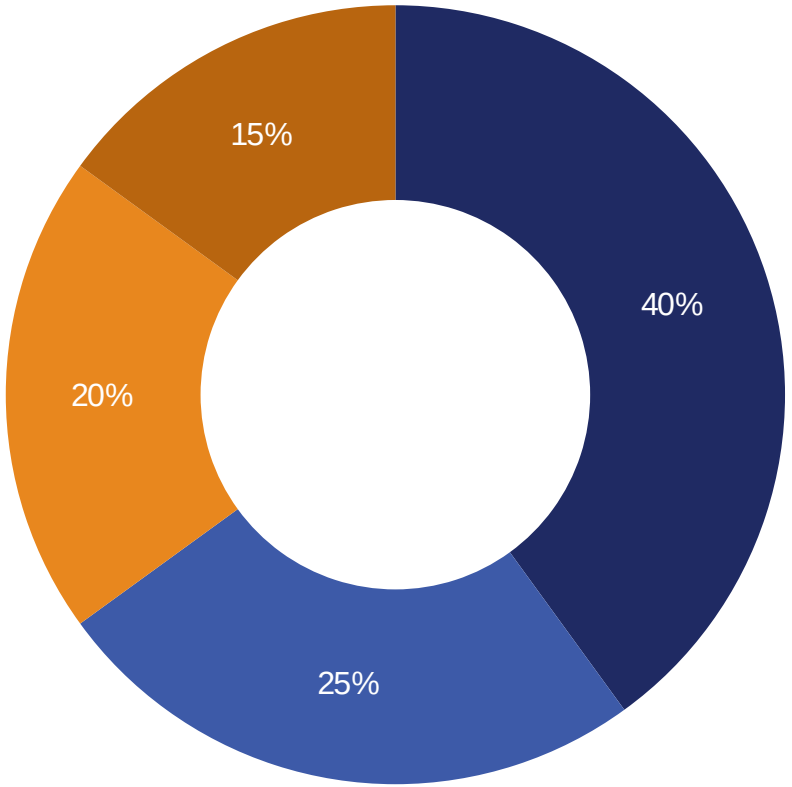
4

NOTIFY

The manager's choice is pushed instantly to the assigned field operator's dashboard.

RISK ENGINE

A transparent, weighted risk score



COMPOSITE SCORE THRESHOLDS

≥ 70 — **HIGH**

Disruption escalated, options re-planned, everyone alerted

≥ 50 — **MEDIUM**

Disruption auto-created with ranked route options

≥ 30 — **AT-RISK**

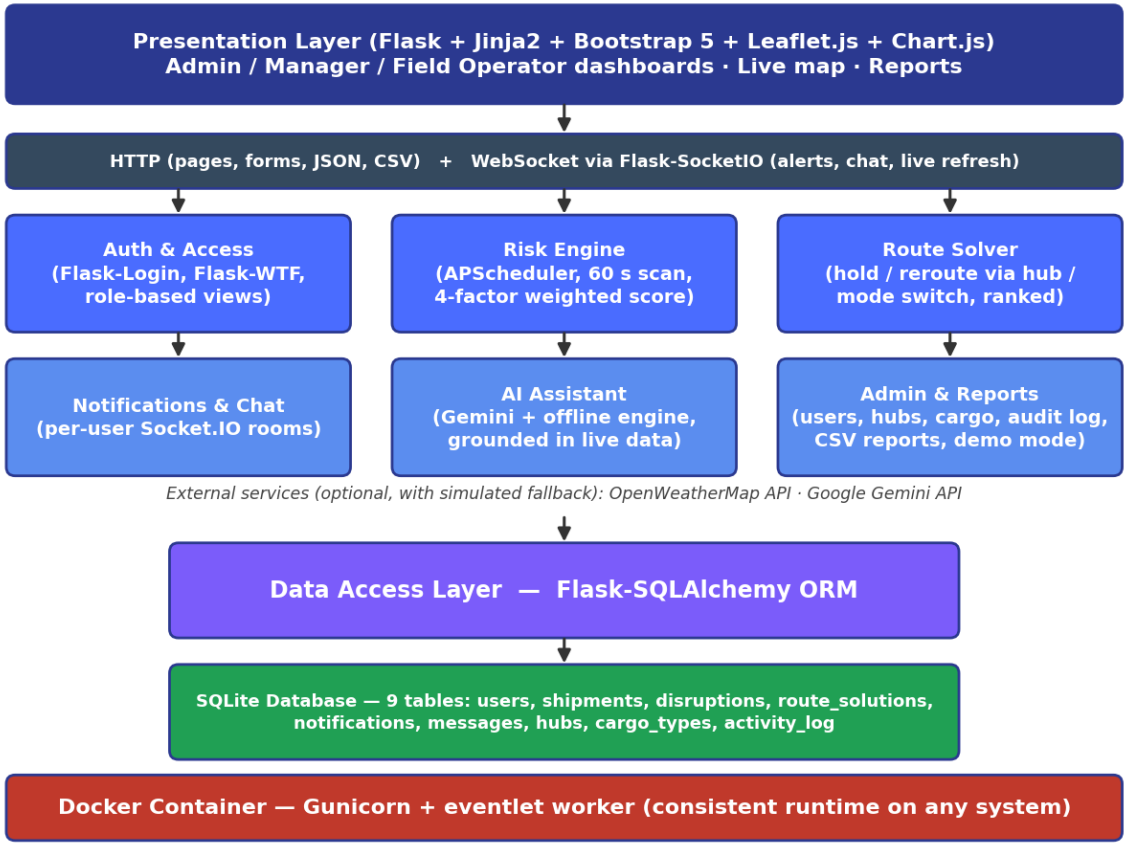
Flagged and watched on dashboards and the live map

< 30 — **LOW**

No action — risk score updated silently

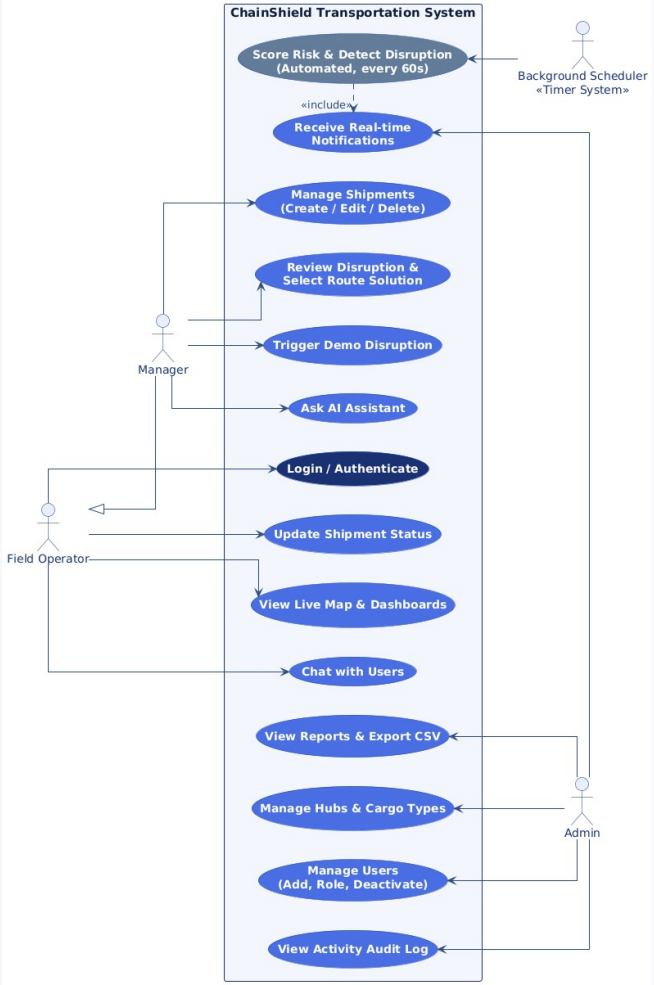
SYSTEM DESIGN

System architecture



SYSTEM DESIGN

Use case diagram



Admin

Manages users, hubs and cargo types; views reports, CSV export and the audit log; receives every alert.

Manager

Creates and edits shipments, reviews disruptions and applies a route solution, triggers the demo, asks the AI assistant. Inherits all Field Operator use cases.

Field Operator

Driver, loco pilot, pilot or captain: logs in, updates shipment status, views the live map and chats with the team.

Background Scheduler

System actor: every 60 s scores risk and detects disruptions, which «include» real-time notifications.

CORE COMPONENTS

What ChainShield does

RE

Risk Engine

Scores every shipment on a recurring schedule

RS

Route Solver

Hold / hub reroute / mode switch, ranked by risk, time & cost

NC

Notify & Chat

Real-time alerts and chat over WebSockets

AP

Admin & Reports

Users, hubs, cargo, audit log and CSV reports

AI

AI Assistant

Gemini Q&A grounded in live data, offline fallback

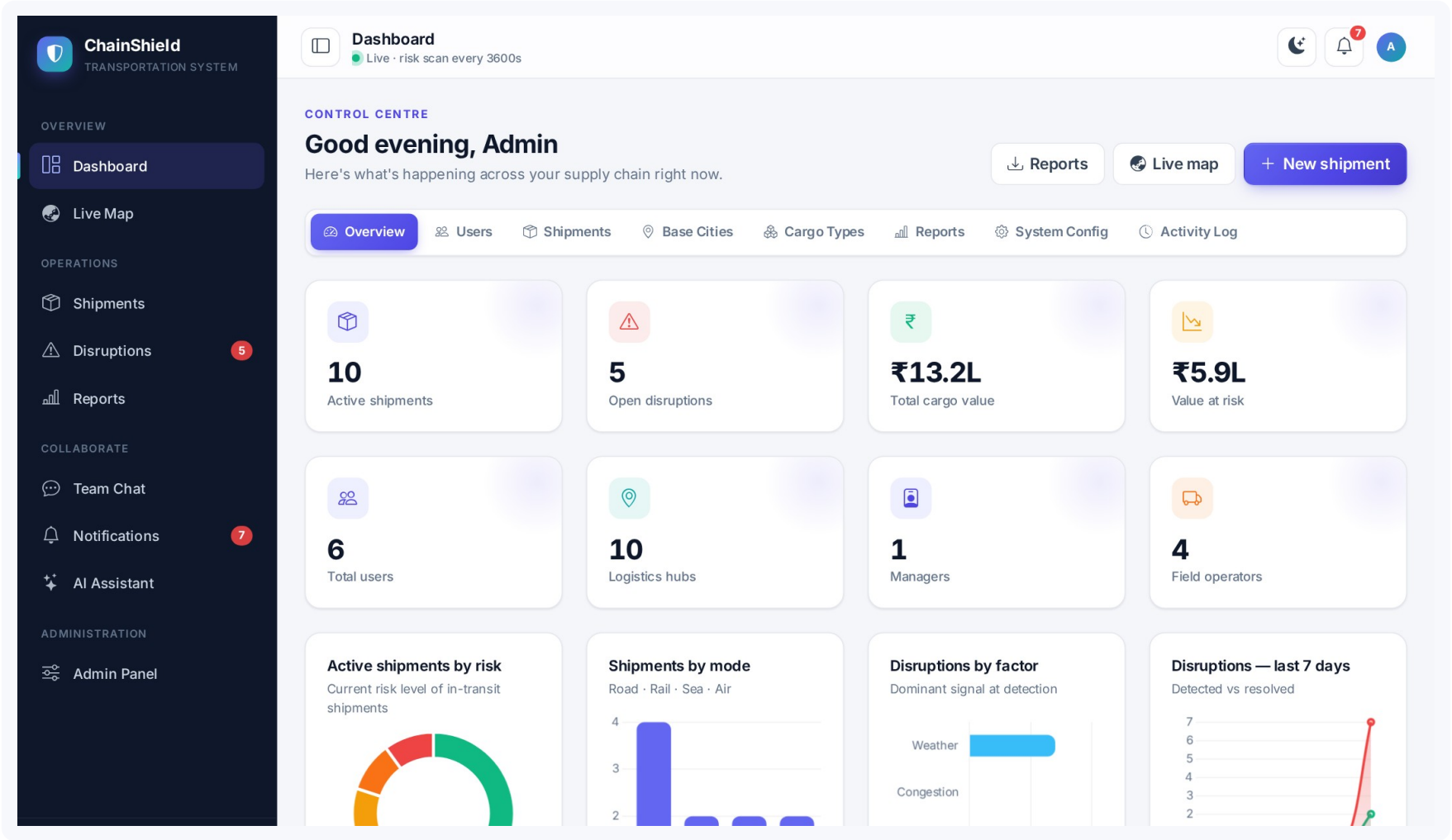
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Live Map

Live positions, routes and disruption zones

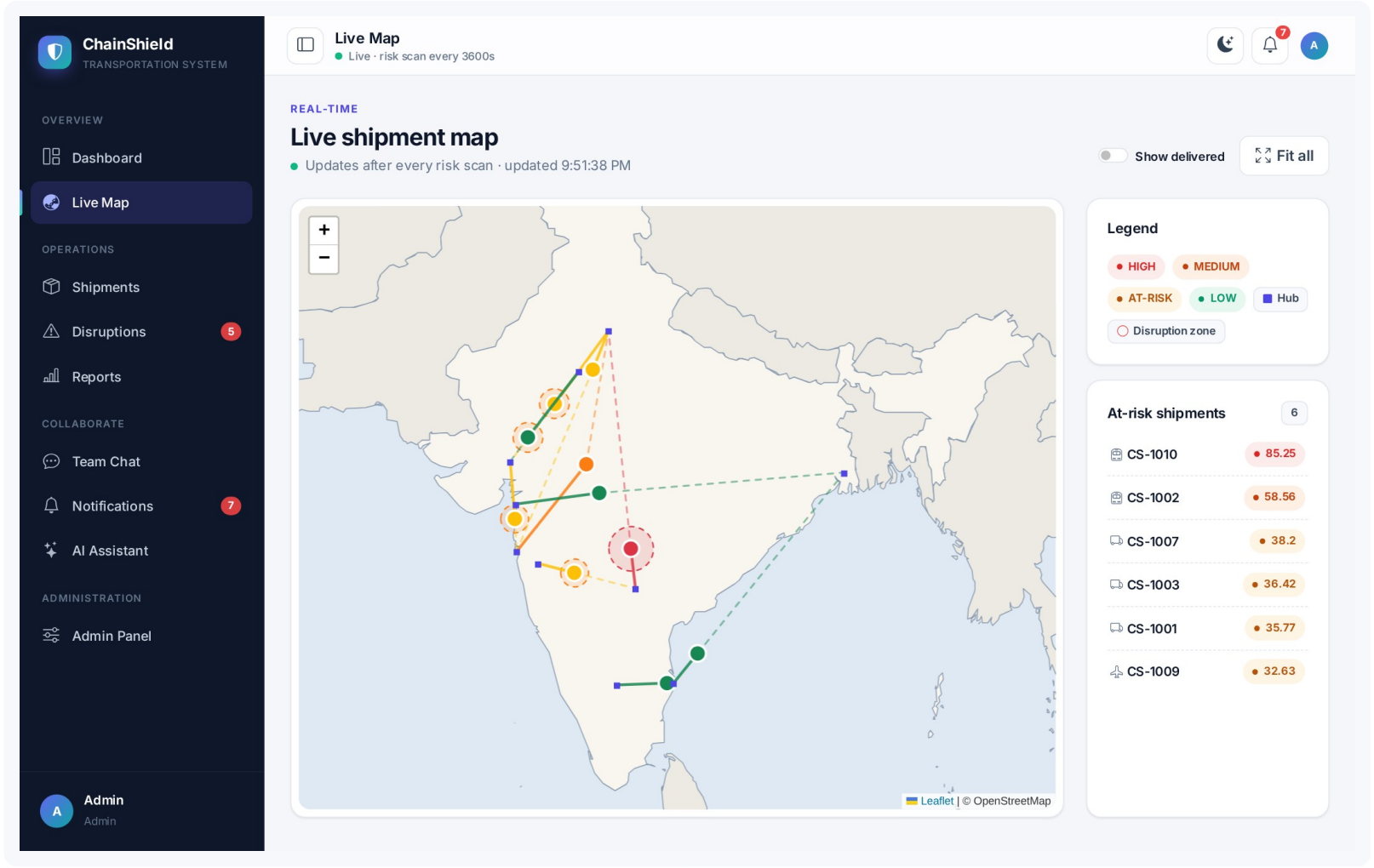
LIVE DEMO

Admin dashboard — live KPIs and charts



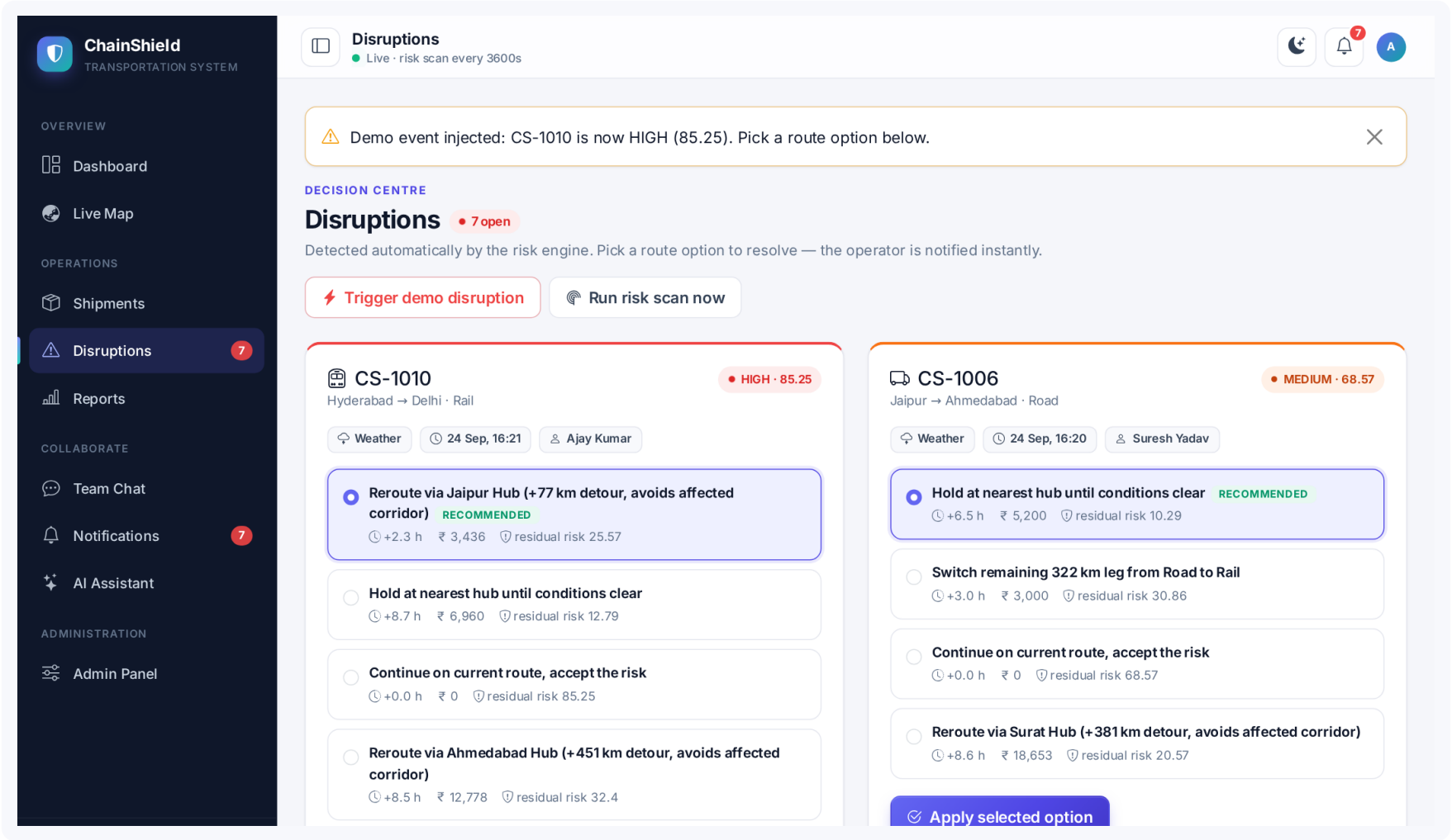
LIVE DEMO

Live shipment map with disruption zones



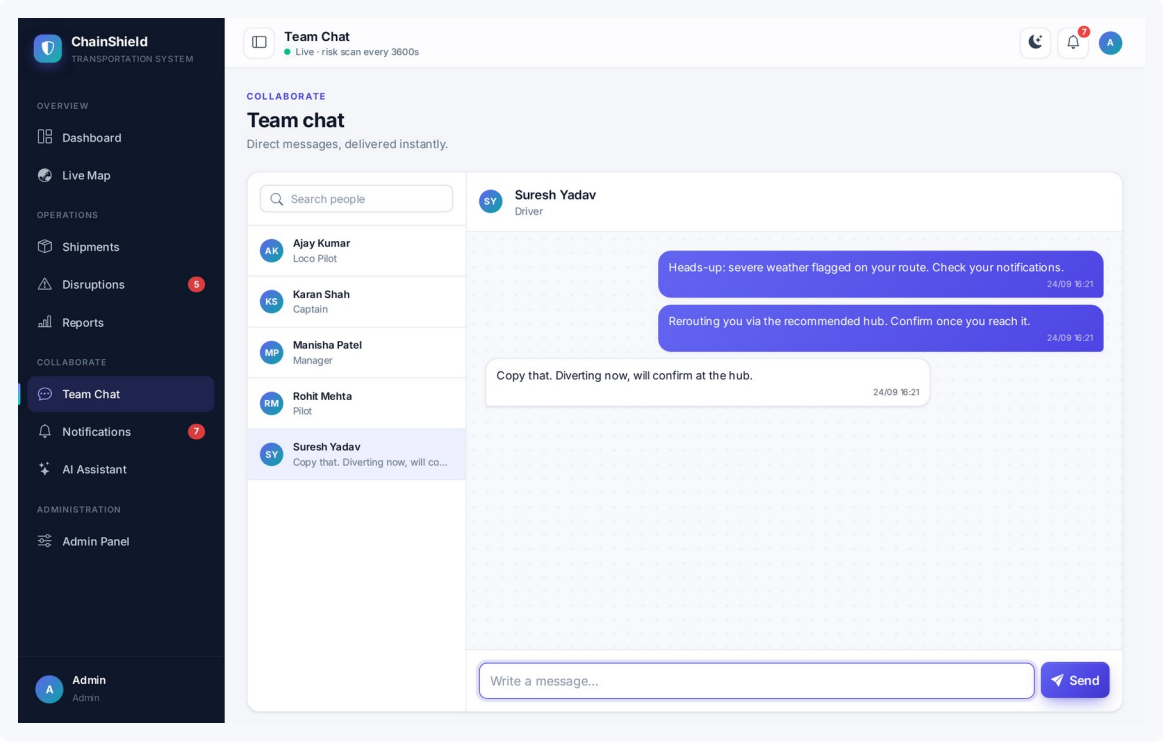
LIVE DEMO

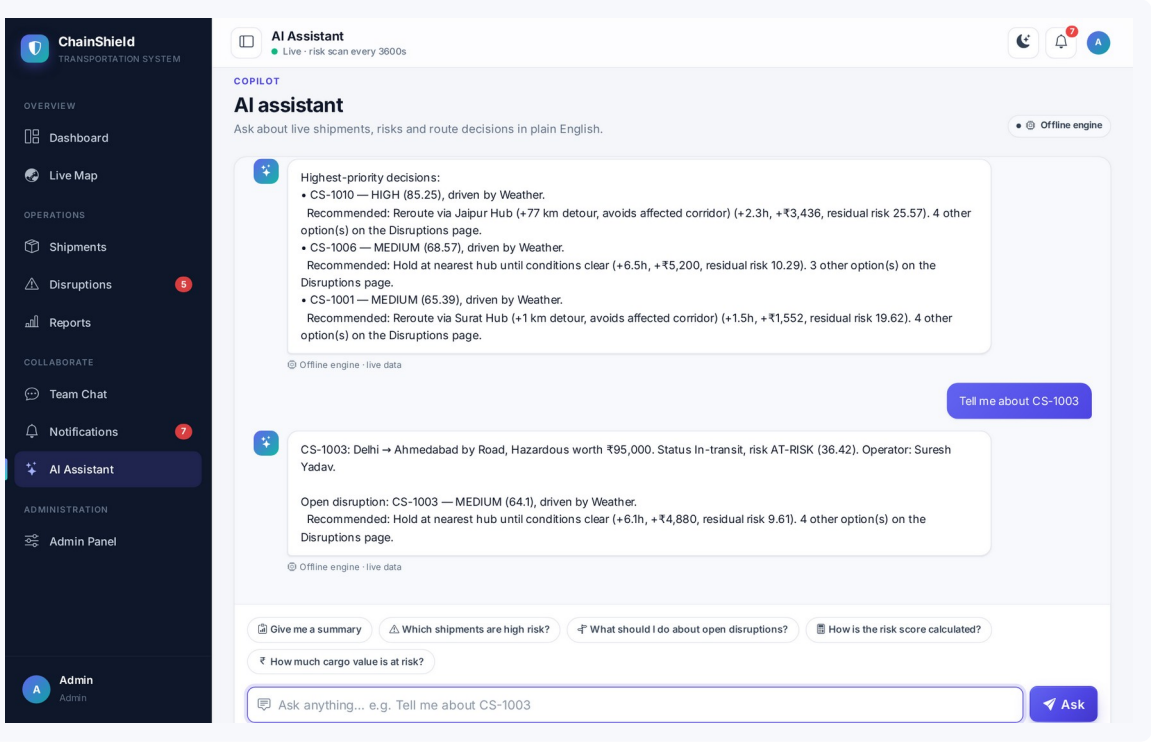
Disruptions with ranked route options



LIVE DEMO

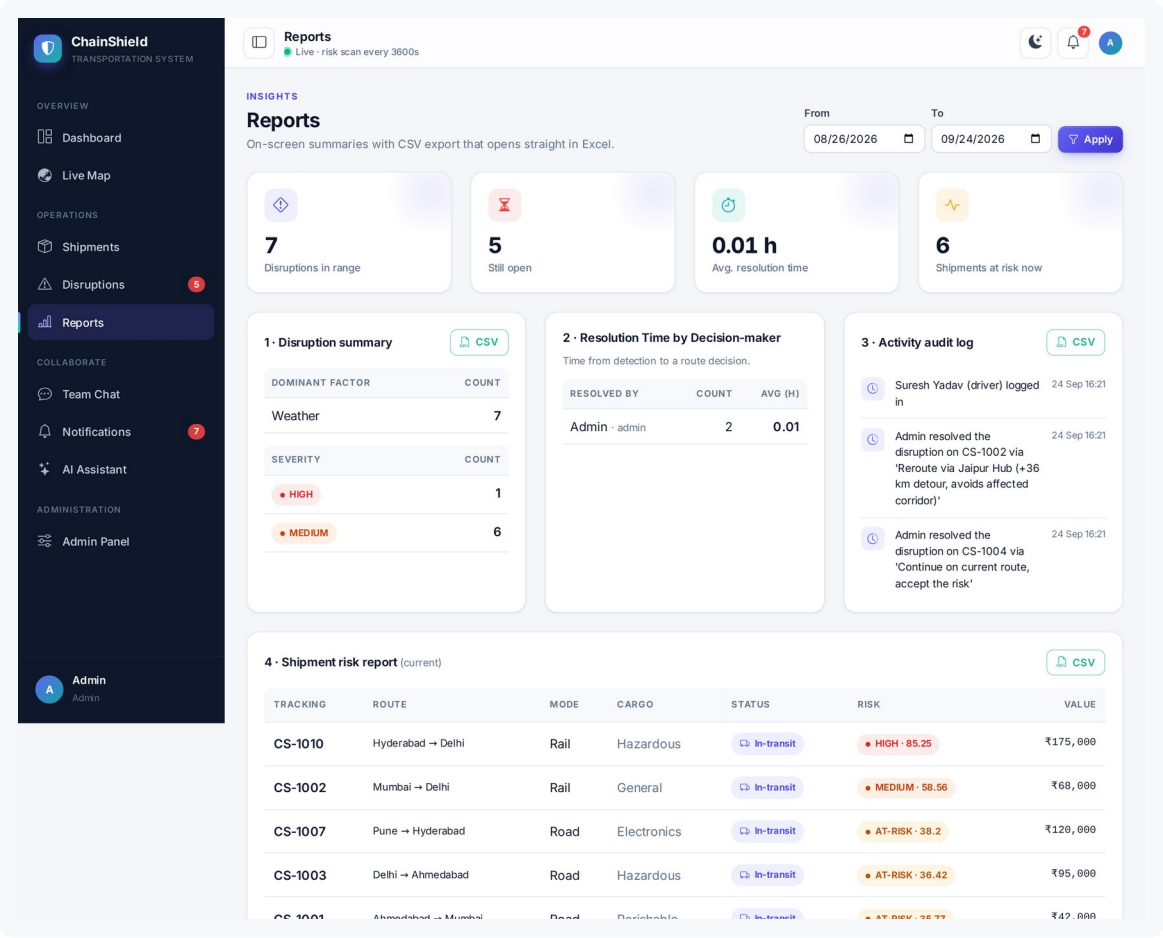
Real-time chat and AI assistant

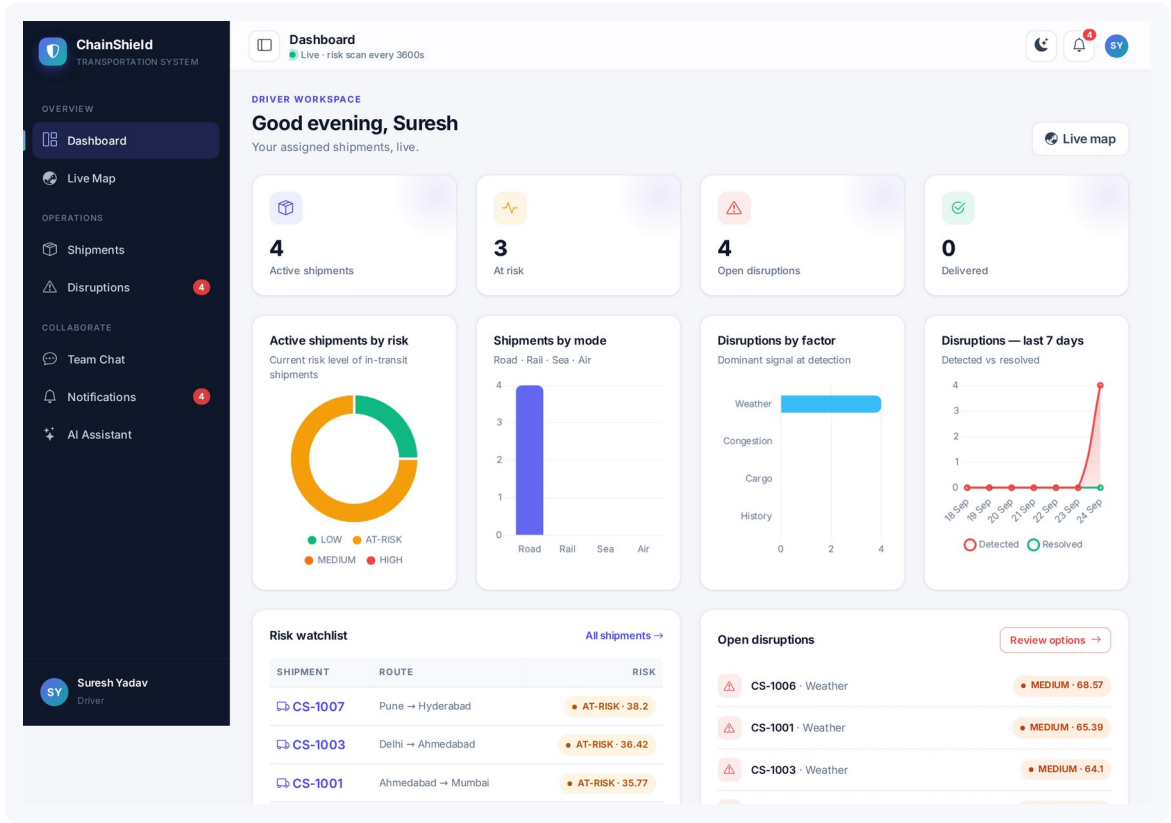




LIVE DEMO

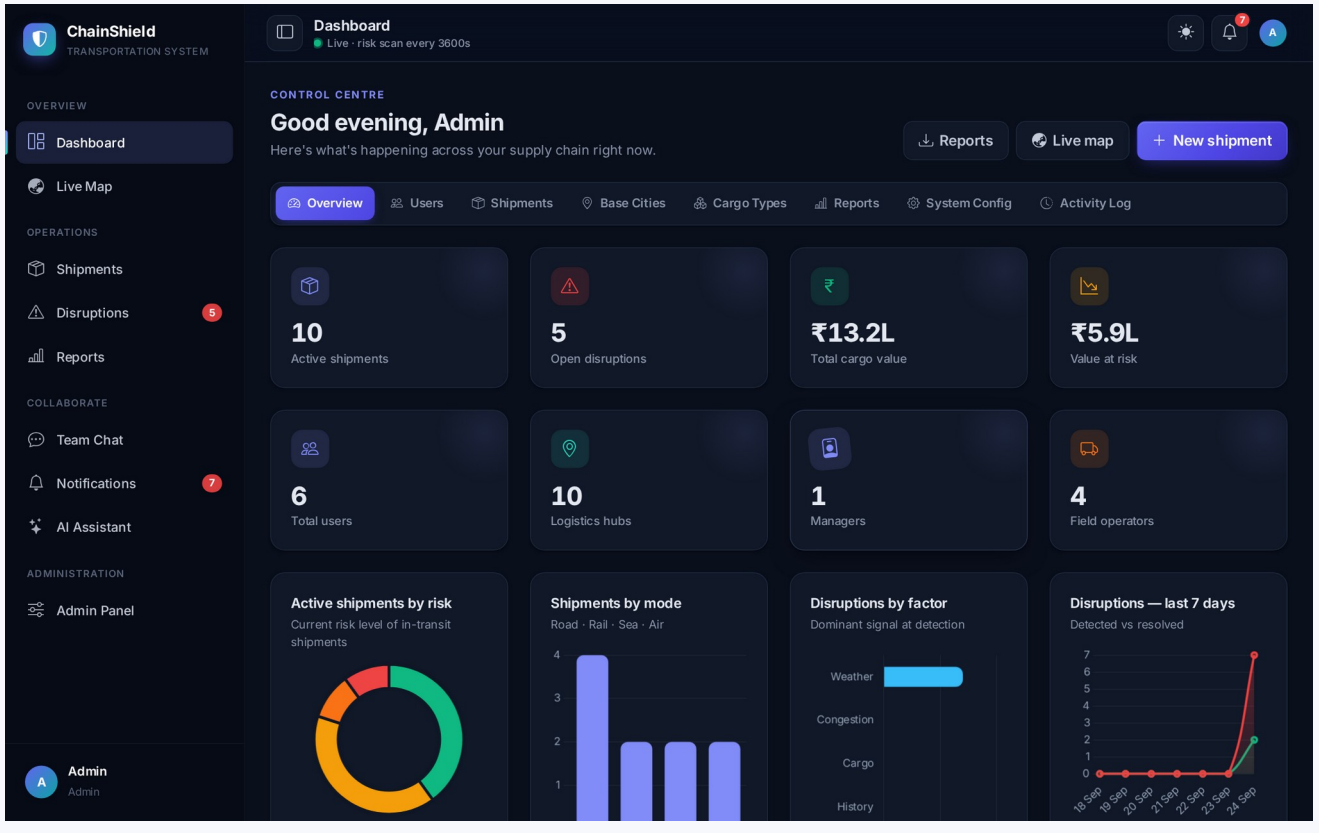
Reports and role-based dashboards

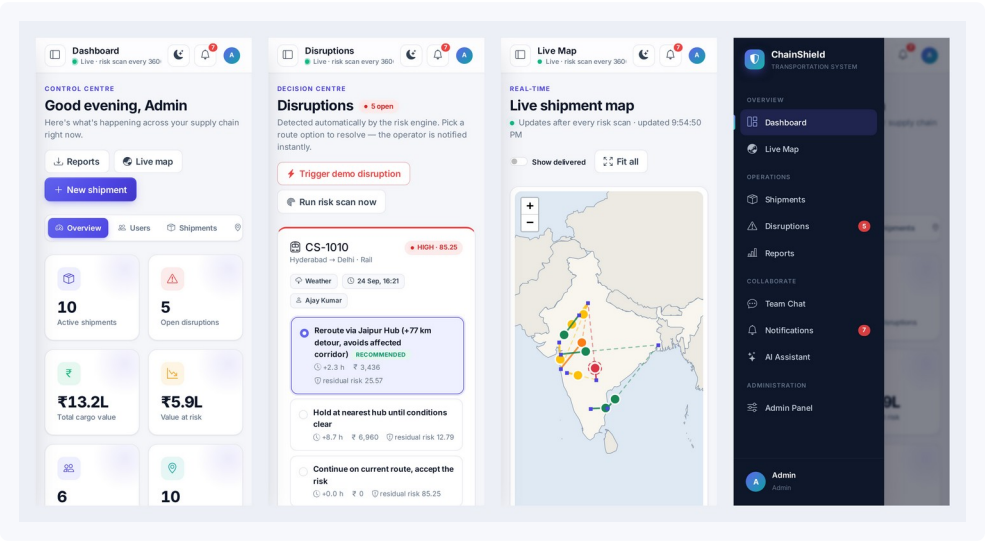




USER EXPERIENCE

Premium, responsive UI — dark mode and mobile





TECHNOLOGY

Technology stack

BACKEND

- Python 3.10
- Flask
- Flask-SQLAlchemy
- Flask-Login
- Flask-SocketIO
- APScheduler

FRONTEND

- Jinja2 Templates
- Bootstrap 5
- JavaScript
- Leaflet.js
- Chart.js

DATA & INFRA

- SQLite
- Docker
- Docker Compose
- Gunicorn + eventlet

EXTERNAL APIS

- Google Gemini API
- OpenWeatherMap API

PROGRESS

Project status — all milestones complete

✓	Requirement Analysis & System Design	DONE
✓	Environment Setup & Docker	DONE
✓	Authentication & Role-based Access	DONE
✓	Risk Engine, Route Solver & Disruptions	DONE
✓	Dashboards, Map, Chat, AI & Premium UI	DONE
✓	Reports, Testing (101 checks) & Docker Deploy	DONE

TESTING

Test results — 101 automated checks, 0 failures

✓	Auth, roles & account deactivation	PASS
✓	Risk formula, disruption creation & escalation	PASS
✓	Route solver ranking & resolution alerts	PASS
✓	Real-time notifications & chat (Socket.IO)	PASS
✓	Reports, CSV export, admin CRUD, demo mode	PASS
✓	Docker build & run (Gunicorn + eventlet)	PASS

CONCLUSION

From reactive tracking to preemptive logistics

ChainShield turns scattered weather, congestion, cargo, and history signals into one transparent risk score per shipment — and pairs every detected disruption with a ranked, actionable route solution, so delays are caught before they cascade.

4	20	101	60s
Weighted risk factors	Functional requirements	Automated checks passed	Risk scan interval

Thank You

Questions & Discussion

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